



1- Traffic Theory

2- Why a new traffic generator ?

3- Results from D-ITG

References

4 - The mean of a Pareto random variable having the above defined parameters is equal to 15, this means that the average time interval between the departure of two consecutive packets is 15 msec; the payload size is 16 bytes. Therefore the average bit rate is equal to the ratio between 128 bits and 15 msec, which yields 8,533Kbps. The output and resulting plots derived from the log file of PCsender confirm our expectations.

4- Using D-ITG in heterogeneous mobile scenario

- The results shown in this poster can be used as reference scenarios for development of wireless communication applications.
- D-ITG enables performance evaluation of **heterogeneous devices** (Laptop, PC desktop, IPAH, ...) over **heterogeneous networks** (Wired LAN, WLAN, ...).

5- Conclusions and Issues for Research

- How to configure next generation networks (NGNs) when more complex traffic bundle rely on it? Extending this topic to include both all "layer 4-7" (VoD, VoIP, FTP, Telnet, HTTP, DNS, SNMP, ...) traffic generation and ICMP could be interesting.
- Presently, our network allows experiments on a small-scale. We would test D-ITG on networks of a much **wider-scale**.
- Mobile Internet access using WLAN and GPRS/3G has gained good popularity. We would test D-ITG in a more large heterogeneous environment made by both heterogeneous (wired and wireless) networks (WLAN, Bluetooth, UMTS, GPRS, GSM...) and heterogeneous users' devices (Laptop, PDA, Advanced Mobile Phone Workstation....).

Acknowledgement

This work has been carried out partially under the financial support of the Ministero dell'Istruzione, dell'Università e della Ricerca (MIUR) in the framework of the FIRB Project "Middleware for advanced services over large-scale, wired-wireless distributed systems (WEB-MINDS)".